Year 12 Further Maths - Complex numbers

Similar questions - Loci, regions and half-lines

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) A locus C_1 is defined by $C_1 = \{z : |z+2\mathrm{i}| \leqslant |z-4|\}$. (i) Indicate by shading on an Argand diagram the region representing C_1. (ii) Find the cartesian equation of the boundary line in the form $ax+by+c=0$. **[4]**

(b) A locus C_2 is defined by $C_2 = \{z : |z+1| \leqslant 3\} \cap \{z : |z-2\mathrm{i}| \geqslant 2\}$. Describe the region representing C_2. **[3]**

2 Two loci are defined by $C_1 = \left\{z : \arg(z+1-\mathrm{i}) = \frac{\pi}{4}\right\}$ and $C_2 = \left\{z : \arg(z-3-\mathrm{i}) = \frac{3\pi}{4}\right\}$. By considering the representations of C_1 and C_2 on an Argand diagram, determine the locus $C_1 \cap C_2$. **[7]**

3

(a) The Argand diagram shows a circle of radius 4 centred on the point representing $3-2\mathrm{i}$. Use set notation to define the locus of complex numbers z on the circle. **[2]**

(b) The locus L is defined by $L = \{z : z \in \mathbb{C},\ |z-2\mathrm{i}| = |z+3|\}$. Sketch and label L. **[2]**

(c) The locus $\left\{z : z \in \mathbb{C},\ \arg(z+1) = \frac{\pi}{3},\ \mathrm{Im}(z) = 3\sqrt{3}\right\}$ contains only one number. Find it. **[3]**

Question 1

(a)(i) $|z+2\mathrm{i}| \leqslant |z-4|$ means z is no further from $-2\mathrm{i}$ (the point $(0,-2)$) than from 4 (the point $(4,0)$): shade the half-plane on the $(0,-2)$ side of the perpendicular bisector (including the line)

(ii) $x^2+(y+2)^2=(x-4)^2+y^2 \Rightarrow 4y+4=-8x+16 \Rightarrow 2x+y-3=0$

(b) Points inside or on the circle with centre $(-1,0)$ and radius 3 and outside or on the circle with centre $(0,2)$ and radius 2 (the circle centred 2i lies partly inside the first circle, so the region is the first disc with the second disc removed)

Question 2

C_1 is the half-line from $-1+\mathrm{i}$ in the direction 45° above the positive real direction: points $-1+\mathrm{i}+t(1+\mathrm{i})$, $t>0$. C_2 is the half-line from $3+\mathrm{i}$ at 135°: $3+\mathrm{i}+s(-1+\mathrm{i})$, $s>0$. Equating: $-1+t=3-s$ and $1+t=1+s$, so $s=t=2$ and the intersection is $1+3\mathrm{i}$. So $C_1 \cap C_2 = \{1+3\mathrm{i}\}$

Question 3

(a) $\{z : |z-(3-2\mathrm{i})|=4\}$ i.e. $\{z : |z-3+2\mathrm{i}|=4\}$

(b) L is the perpendicular bisector of the points 2i and -3: the line through the midpoint $(-1.5,1)$ with gradient $-\frac{3}{2}$, i.e. $6x+4y+5=0$

(c) $z+1=r\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)$ with $\mathrm{Im}=\frac{\sqrt{3}}{2}r=3\sqrt{3} \Rightarrow r=6$; $z+1=3+3\sqrt{3}\mathrm{i}$, so $z=2+3\sqrt{3}\mathrm{i}$